

EXPERIMENT - 17

Title

Create a React Application for an Online Store with Registration, Login, Product Pages and Routing.

Aim

To develop a fully functional Online Store application using:

- ReactJS
 - React Router DOM
 - Component-based architecture
 - Modern UI styling
 - Navigation between pages
-

STEP 1 - Create Project

Shell

```
npx create-react-app online-store  
cd online-store  
npm install react-router-dom  
npm start
```

Final Folder Structure

None

```
online-store/  
|  
├─ src/  
|   ├─ components/  
|   |   ├─ Navbar.jsx  
|   |   └─ ProductCard.jsx  
|   |  
|   └─ pages/  
|       ├─ Home.jsx  
|       ├─ Register.jsx  
|       ├─ Login.jsx  
|       ├─ Products.jsx  
|       └─ ProductDetails.jsx  
|  
├─ App.jsx  
├─ App.css  
└─ index.js
```

index.js

JavaScript

```
import React from "react";  
import ReactDOM from "react-dom/client";  
import App from "./App";  
import "./App.css";  
  
const root =  
  ReactDOM.createRoot(document.getElementById("root"));
```

```
root.render(<App />);
```

App.jsx (Routing Setup)

JavaScript

```
import React from "react";
import { BrowserRouter as Router, Routes, Route } from
"react-router-dom";
import Navbar from "../components/Navbar";
import Home from "../pages/Home";
import Register from "../pages/Register";
import Login from "../pages/Login";
import Products from "../pages/Products";
import ProductDetails from "../pages/ProductDetails";

function App() {
  return (
    <Router>
      <Navbar />
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/register" element={<Register />} />
        <Route path="/login" element={<Login />} />
        <Route path="/products" element={<Products />} />
        <Route path="/products/:id"
element={<ProductDetails />} />
      </Routes>
    </Router>
  );
}
```

```
export default App;
```

Navbar.jsx

JavaScript

```
import React from "react";
import { Link } from "react-router-dom";

function Navbar() {
  return (
    <nav className="navbar">
      <h2> Online Store</h2>
      <div>
        <Link to="/">Home</Link>
        <Link to="/products">Products</Link>
        <Link to="/login">Login</Link>
        <Link to="/register">Register</Link>
      </div>
    </nav>
  );
}

export default Navbar;
```

Home.jsx

JavaScript

```
import React from "react";

function Home() {
  return (
    <div className="page">
      <h1>Welcome to Our Online Store</h1>
      <p>Shop the best products with premium quality.</p>
    </div>
  );
}

export default Home;
```

Register.jsx

JavaScript

```
import React, { useState } from "react";

function Register() {
  const [user, setUser] = useState({});
  const [message, setMessage] = useState("");

  const handleSubmit = (e) => {
    e.preventDefault();
    setMessage("Registration Successful!");
  };

  return (
    <div className="form-container">
      <h2>Register</h2>
```

```

    <form onSubmit={handleSubmit}>
      <input type="text" placeholder="Full Name"
        onChange={(e) => setUser({...user, name:
e.target.value})} required />
      <input type="email" placeholder="Email"
        onChange={(e) => setUser({...user, email:
e.target.value})} required />
      <input type="password" placeholder="Password"
        onChange={(e) => setUser({...user, password:
e.target.value})} required />
      <button type="submit">Register</button>
    </form>
    <p className="success">{message}</p>
  </div>
);
}

export default Register;

```

Login.jsx

JavaScript

```

import React, { useState } from "react";

function Login() {
  const [message, setMessage] = useState("");

  const handleLogin = (e) => {
    e.preventDefault();
    setMessage("Login Successful!");
  };
}

```

```
};

return (
  <div className="form-container">
    <h2>Login</h2>
    <form onSubmit={handleLogin}>
      <input type="email" placeholder="Email" required />
      <input type="password" placeholder="Password"
required />
      <button type="submit">Login</button>
    </form>
    <p className="success">{message}</p>
  </div>
);
}

export default Login;
```

Products.jsx

JavaScript

```
import React from "react";
import { Link } from "react-router-dom";
import ProductCard from "../components/ProductCard";

const productList = [
  { id: 1, name: "Laptop", price: 60000 },
  { id: 2, name: "Smartphone", price: 25000 },
  { id: 3, name: "Headphones", price: 3000 }
];
```

```
function Products() {
  return (
    <div className="products">
      {productList.map(product => (
        <Link key={product.id}
to={` /products/${product.id}`}>
        <ProductCard product={product} />
      </Link>
      )}}
    </div>
  );
}

export default Products;
```

ProductCard.jsx

```
JavaScript
import React from "react";

function ProductCard({ product }) {
  return (
    <div className="card">
      <h3>{product.name}</h3>
      <p>₹{product.price}</p>
    </div>
  );
}
```



```
export default ProductCard;
```

ProductDetails.jsx

JavaScript

```
import React from "react";
import { useParams } from "react-router-dom";

const products = [
  { id: 1, name: "Laptop", price: 60000, desc: "High performance laptop" },
  { id: 2, name: "Smartphone", price: 25000, desc: "Latest Android smartphone" },
  { id: 3, name: "Headphones", price: 3000, desc: "Noise cancelling headphones" }
];

function ProductDetails() {
  const { id } = useParams();
  const product = products.find(p => p.id === parseInt(id));

  return (
    <div className="page">
      <h2>{product.name}</h2>
      <p>{product.desc}</p>
      <h3>Price: ₹{product.price}</h3>
    </div>
  );
}
```

```
export default ProductDetails;
```

App.css

CSS

```
body {  
  margin: 0;  
  font-family: Arial, sans-serif;  
  background: linear-gradient(to right, #141e30, #243b55);  
  color: white;  
}  
  
.navbar {  
  display: flex;  
  justify-content: space-between;  
  padding: 15px 40px;  
  background: rgba(0,0,0,0.8);  
}  
  
.navbar a {  
  margin-left: 20px;  
  color: white;  
  text-decoration: none;  
  font-weight: bold;  
}  
  
.page {  
  text-align: center;  
  margin-top: 60px;
```

```
}

.products {
  display: flex;
  justify-content: center;
  gap: 30px;
  margin-top: 50px;
}

.card {
  background: white;
  color: black;
  padding: 20px;
  border-radius: 10px;
  width: 200px;
  transition: 0.3s;
}

.card:hover {
  transform: scale(1.05);
}

.form-container {
  width: 300px;
  margin: 50px auto;
  background: white;
  color: black;
  padding: 25px;
  border-radius: 10px;
}

input {
  width: 100%;
```

```
padding: 8px;
margin: 8px 0;
}

button {
width: 100%;
padding: 8px;
background: #243b55;
color: white;
border: none;
cursor: pointer;
}

.success {
color: green;
font-weight: bold;
}
```

Final Result

This Online Store Application includes:

- ✓ Registration Page
- ✓ Login Page
- ✓ Product Listing
- ✓ Product Details
- ✓ Routing
- ✓ Component Architecture
- ✓ Professional Styling
- ✓ Dynamic Navigation

